

ENTERPRISE SYSTEM ARCHITECTURE, TECHNOLOGY STACK & BACKEND SPECIFICATION

Statutory ESG & SEBI BRSR Core Digital Engine for MEIL Group (250+ Project Sites)

This technical document provides the complete, authoritative engineering specification of the MEIL ESG / BRSR Enterprise Portal. It outlines the end-to-end multi-tier system architecture, comprehensive technology stack choices, relational data models, RESTful API catalog, internal micro-engines (GHG Protocol Scope 1/2/3, CEA Indian Grid v19 factors, NGRBC 9 principles), immutable evidence vault, strict Role-Based Access Control (RBAC), and cryptographic audit assurance mechanisms designed to satisfy SEBI Circular (Jan 2025) and ICAI Revised 2024 assurance protocols.

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Target Entity	Megha Engineering & Infrastructures Ltd. (CIN: U45202TG2006PLC050271)
Turnover Scope	INR 32,450+ Crores (Top 1,000 Listed Entity Scope)
Mandatory Standards	SEBI BRSR (May 2021), BRSR Core (July 2023 / Jan 2025), NGRBC 9 Principles
Assurance Protocol	ICAI Reasonable Assurance Standard (Revised 2024)
Effective Date	October 10, 2026
Architecture Status	PRODUCTION READY & DEPLOYED

DOCUMENT CONTENTS AT A GLANCE

- Executive Summary & Enterprise Operational Scope — Regulatory Mandates & Operational Challenges
- End-to-End System Topology & High-Level Architecture — 4-Tier Enterprise Topology & Ingestion
- Full-Stack Technology Matrix — React 19, FastAPI 0.110, SQLAlchemy 2.0, ReportLab, Pydantic v2
- Deep-Dive Backend Architecture & Core Calculation Engines — GHG Scope 1/2/3, CEA Grid, 4-Eye Workflow
- Complete API Route Inventory & RESTful Endpoints Catalog — All 16 Modular Namespace Routers
- Relational Database Schema & Data Models Specification — 38 Relational Models & Audit Tables
- Role-Based Access Control (RBAC) & Security Architecture — Dual-Vector Permission Matrix & JWT
- Evidence Vault, SHA-256 Immutability & ICAI Audit Trail — Zero-Tamper Chain of Custody
- Deployment Architecture, Operational Runbook & Production Verification — Alembic, Testing & Uvicorn

1. Executive Summary & Enterprise Operational Scope

Megha Engineering & Infrastructures Limited (MEIL) is one of India's preeminent engineering conglomerates, managing over 250 mega-infrastructure projects across Hydrocarbons, Renewable Energy, Thermal Power, Irrigation & Lift Drinking Water Systems, High-Altitude Tunnels (e.g., Zojila Pass), Electric Mobility (Olectra), and Urban Mass Rapid Transit. Due to its market capitalization, scale of operations, and capital structure, MEIL falls under the statutory jurisdiction of the **Securities and Exchange Board of India (SEBI) BRSR Mandate**.

1.1 Regulatory Compliance Mandates

The MEIL ESG Enterprise Portal was custom-engineered to address three non-negotiable compliance pillars:

- **SEBI Circular May 2021 (BRSR Framework):** Comprehensive disclosure across General Disclosures (Section A), Management & Process Disclosures (Section B), and Principle-wise Performance Disclosures (Section C) encompassing National Guidelines on Responsible Business Conduct (NGRBC) Principles 1 through 9.
- **SEBI Circular July 2023 & Jan 2025 (BRSR Core):** Mandatory Reasonable Assurance on 9 designated ESG KPI attributes (Greenhouse Gas Footprint, Water Intensity, Energy Consumption, Waste Generation & Circularity, Employee Wellbeing, Gender Diversity, POSH & Safety, Inclusive Development, and Fair Supply Chains).
- **ICAI Revised Assurance Standard (2024):** Mandating cryptographic immutability, source document linking, tamper-evident audit trails, and strict 4-Eye authorization workflow for every ESG metric.

1.2 Enterprise Architectural Challenges Solved

Challenge	Operational Context	Architectural Solution in MEIL Portal
Decentralized Project Sites	250+ active construction & infrastructure locations spread nationally with variable connectivity.	Hierarchical data ingestion tree (Project → Business Unit → Subsidiary → Group HQ) with local caching.
Audit Defense & Greenwashing Risks	Statutory penalties under Companies Act & SEBI Regulations for unsubstantiated ESG claims.	Evidence Vault enforcing mandatory SHA-256 file hashing, file locking, and direct citation linking.
Grid Emission Variability	Electrical consumption across multiple state distribution utilities with differing carbon factors.	Dynamic Emission Engine bound to official Central Electricity Authority (CEA) Indian Grid Baseline Database v19.
Departmental Silos	Data spread across Site Engineering, HSE, HR, Procurement, Legal & CSR departments.	Strict Role-Based Access Control (RBAC) preventing cross-portal leakage while consolidating to Group HQ.

2. End-to-End System Topology & Architecture

The MEIL ESG platform employs a modern **Decoupled Asynchronous Micro-Layered Architecture**. The architecture guarantees sub-100ms response times for telemetry aggregation, multi-layer cryptographic validation, and zero-latency reporting periods across 4 distinct operational tiers:

TIER 1 — PRESENTATION & WORKSPACE INTERFACE (React 19 + Vite 8 SPA)

- **Executive HQ Dashboard:** Multi-subsidiary ESG cockpit, Scope 1/2/3 charts, BRSR readiness gauges.
- **Site Operations Module:** Rapid mobile/desktop entry for fuel, energy, water, waste, and concrete logs.
- **Evidence Vault Manager:** Drag-and-drop document upload with client-side SHA-256 pre-validation.
- **Auditor Review & Verification Portal:** 1-click sample inspection, audit trail viewer, certificate export.

▼ *Encrypted HTTPS / RESTful JSON / Bearer JWT Authorization Headers (HS256/RS256)* ▼

TIER 2 — API GATEWAY & SECURITY CONTROLS (FastAPI ASGI Perimeter)

- **CORS & Host Whitelisting:** Restricts cross-origin requests to authorized enterprise domains.
- **OAuth2 & JWT Validator:** Resolves token claims, validates expiration, checks token blocklist.
- **Dual-Vector RBAC Filter:** Enforces user role permissions × organizational boundary (Project / BU / Sub / Group HQ).
- **Request Audit Interceptor:** Logs client IP, user ID, endpoint route, and timestamp for all mutating calls.

▼ *Stateless Python Inversion of Control (FastAPI Dependency Injection get_db)* ▼

TIER 3 — BUSINESS LOGIC & SPECIALIZED MICRO-ENGINES (app/services)

- **Emission Engine:** GHG Scope 1 (Direct Fuel), Scope 2 (CEA Indian Grid v19), Scope 3 Value Chain.
- **Consolidation Engine:** 4-level enterprise tree rollup with turnover-weighted ESG intensities.
- **Validation Engine:** Month-on-month ±30% anomaly detection, physics rules, required evidence check.
- **4-Eye Workflow Engine:** Enforces strict state machine: DRAFT → SUBMITTED → UNDER_REVIEW → APPROVED → LOCKED.
- **BRSR Core Engine:** Maps operations into SEBI NGRBC Principles 1-9 & 9 Core Reasonable Assurance attributes.
- **Report Generator:** PyMuPDF (fitz), ReportLab 5.0, and openpyxl statutory artifact generation with SHA-256 seal.

▼ *SQLAlchemy 2.0 ORM Transaction Lifecycle (Unit of Work Pattern / ACID Guaranteed)* ▼

TIER 4 — DATA PERSISTENCE & IMMUTABLE ARTIFACT VAULT (Storage Layer)

- **Relational Database (SQLite / PostgreSQL):** 38 tables with foreign-key referential integrity & Alembic migrations.
- **Immutable Evidence Storage:** Dedicated /storage/evidence/ repository with enforced SHA-256 integrity checks.
- **Issued Reports Archive:** Dedicated /reports_storage/ housing permanent PDF/XLSX statutory disclosures.

3. Full-Stack Technology Matrix

Every technology component was chosen based on benchmarked performance, memory footprint, security hardening, and enterprise sustainability. Below is the complete technological inventory:

Component Layer	Technology / Library	Version	Architectural Justification
Frontend Framework	React (SPA)	19.0.0	Component modularity, concurrent rendering, virtual DOM state synchronization.
Build Tooling	Vite	8.3.2	Sub-second Hot Module Replacement (HMR), tree-shaking, lightweight Rollup bundling.
Icons & Visuals	Lucide React	0.344.0	Zero-runtime SVG rendering, accessible iconography across enterprise dashboards.
Client PDF Render	html2canvas + jsPDF	1.4.1 / 2.5.1	Client-side zero-latency audit certificates and instant snapshot downloads.
XSS Defense	DOMPurify	3.0.9	Strict sanitization of user-submitted notes and auditor commentary.
Backend Web API	FastAPI (ASGI)	0.110.0+	Asynchronous execution, Starlette performance, automatic OpenAPI / Swagger spec generation.
ASGI Server	Uvicorn	0.28.0+	High-throughput lightning-fast ASGI server with auto-reload and worker clustering.
Data Validation	Pydantic v2	2.6.0+	Rust-backed validation core, strict typing, schema serialization, payload filtering.
ORM Engine	SQLAlchemy 2.0	2.0.28+	Modern `select()` syntax, typed models, session scoping, enterprise transaction isolation.
Database Schema Migrations	Alembic	1.13.0+	Zero-downtime declarative database migrations with deterministic schema revision hashes.
Relational Store	SQLite 3 / PostgreSQL	3.45 / 16.0	ACID compliance, WAL mode (Write-Ahead Logging), high-concurrency read efficiency.
Security & Auth	python-jose + passlib	3.3.0 / 1.7.4	JWT signed tokens (HS256/RS256), bcrypt password hashing with salt rounds ≥ 12 .
Server PDF Engine	PyMuPDF (fitz) + ReportLab	1.24.0 / 5.0.1	High-speed programmatic statutory document generation, vector drawings, exact SEBI layout.
Spreadsheet Engine	openpyxl	3.1.2+	Direct creation of multi-tab SEBI BRSR Excel workbooks with formula binding.
Testing & QA	Pytest + HTTPX	8.0.0 / 0.27.0	Full API regression testing suite, async test client, mathematical validation checks.

4. Deep-Dive Backend Architecture & Core Calculation Engines

The backend is encapsulated within the `app/services/` module, providing deterministic, stateless, and mathematically auditable micro-engines. Each engine operates strictly under the transaction boundaries of SQLAlchemy sessions.

4.1 Emission Calculation Engine ([app/services/emission_engine.py](#))

The Emission Engine calculates GHG Protocol Scope 1, Scope 2, and Scope 3 emissions using regulatory factor tables.

- **Scope 1 Direct Emissions:** Computes CO₂, CH₄, and N₂O emissions from stationary combustion (DG sets, heavy boilers), mobile equipment (excavators, dump trucks), and process flaring using stoichiometric emission factors (e.g., High-Speed Diesel = 2.68 kg CO₂e / Litre).
- **Scope 2 Indirect Emissions (Electricity):** Implements the CEA (Central Electricity Authority, Ministry of Power, GoI) Indian Grid Baseline Emission Factor v19 (0.716 kg CO₂e / kWh). Supports location-based and market-based accounting.
- **Scope 3 Value Chain Emissions:** Aggregates upstream supply chain transport, business travel, waste disposal, and contractor equipment operations.
- **Unit Normalization Engine:** Automatically converts Litres, Barrels, Gallons, MT, kWh, MWh, and GJ into standard SI metric equivalents before applying factors.

4.2 Group Consolidation Engine ([app/services/consolidation_engine.py](#))

Consolidates operational data across the 4-level enterprise tree: **Project** → **Business Unit** → **Subsidiary** → **Group HQ**.

- **Consolidation Approach:** Adheres to the Financial Control / Operational Control criteria defined by GHG Protocol and SEBI Circular.
- **Rollup Aggregation:** Performs recursive rollup of fuel, energy, water, waste, and workforce numbers with weighted-average intensity metrics.
- **Intensity Metrics Computation:** Automatically computes BRSR Core intensities per Crore INR turnover and per physical infrastructure output.

4.3 Validation & Anomaly Detection Engine ([app/services/validation_engine.py](#))

Provides automated pre-submission data hygiene, fraud detection, and sanity verification.

- **Tolerance Band Checking:** Compares month-on-month variance. Variations > ±30% trigger an automatic WARNING requiring mandatory officer explanation.
- **Physics-Based Rules:** Validates that water discharge ≤ water consumption, diesel fuel hours correlate with equipment operational logbooks.
- **Evidence Mandatory Rule:** If an expenditure or resource exceeds the statutory threshold, submission without an attached evidence document is blocked.

4.4 Workflow & 4-Eye Approval Engine ([app/services/workflow_engine.py](#))

Manages the statutory state machine governing reporting periods and submission packages.

- **State Machine Lifecycle:** DRAFT → SUBMITTED → UNDER_REVIEW → APPROVED (or REJECTED) → LOCKED.
- **4-Eye Principle:** The user who inputs the data (Site Officer) is cryptographically prevented from approving their own package. Approval requires designated BU Lead or Group Sustainability Auditor.
- **Period Freezing:** Once approved, reporting periods transition to LOCKED, permanently freezing database records against retroactive mutation.

4.5 BRSR Framework & Readiness Engine (app/services/brsr_engine.py)

Maps low-level operational records into statutory SEBI BRSR Section A, B, and C indicator disclosures.

- **NGRBC 9 Principles:** Auto-populates indicators across Ethics (P1), Sustainable Goods (P2), Employee Wellbeing (P3), Stakeholder Engagement (P4), Human Rights (P5), Environment (P6), Public Policy (P7), Inclusive Growth (P8), and Customer Value (P9).
- **Readiness Scoring:** Computes weighted readiness percentages for Essential vs. Leadership indicators.
- **BRSR Core Reasonable Assurance Score:** Tracks audit readiness of designated Core attributes.

4.6 Tamper-Evident Audit Logging Engine (app/services/audit_service.py)

Implements an immutable ledger capturing every state change, document upload, and user action.

- **Log Metadata:** Records timestamp (UTC), User ID, User Role, Action Code, Target Entity, Old Values (JSON), New Values (JSON), and Client IP address.
- **Query Interface:** Provides paginated, filtered audit trails for external statutory assurance teams (e.g., PwC, EY, KPMG, BDO).

5. Complete API Route Inventory & REST Endpoints Catalog

The backend exposes **16 dedicated modular routers** under the ``/api/v1`` namespace. Every endpoint enforces strict Pydantic input/output schemas and OAuth2/JWT scope verification:

Router Namespace	Method & Primary Endpoints	RBAC Roles Allowed	Functional Responsibility
auth.py <code>`/api/v1/auth`</code>	<code>`POST /login`</code> <code>`POST /logout`</code> <code>`GET /me`</code> <code>`POST /refresh`</code>	Public / All Authenticated	Issues JWT access tokens, validates credentials against bcrypt hash, maintains token blocklist.
organization.py <code>`/api/v1/organization`</code>	<code>`GET /hierarchy`</code> <code>`GET /projects`</code> <code>`POST /projects`</code> <code>`GET /projects/{id}`</code>	All Roles (Filtered by Scope)	Maintains Group HQ, Subsidiaries, Business Units, and 250+ project site nodes.
reporting_periods.py <code>`/api/v1/reporting-periods`</code>	<code>`GET /`</code> <code>`POST /`</code> <code>`POST /{id}/lock`</code> <code>`GET /active`</code>	Sustainability Head, Auditor	Defines financial years (e.g. FY 2024-25), quarter cycles, and lock states.
submissions.py <code>`/api/v1/submissions`</code>	<code>`GET /`</code> <code>`POST /`</code> <code>`POST /{id}/transition`</code> <code>`GET /{id}/audit-trail`</code>	Site Officer, BU Lead, Auditor	Manages departmental submission packages and 4-Eye approval state machine.
esg_records.py <code>`/api/v1/esg-records`</code>	<code>`GET /fuel`</code> <code>`POST /fuel`</code> <code>`GET /energy`</code> <code>`POST /energy`</code> <code>`GET /water`</code> <code>`POST /water`</code> <code>`GET /waste`</code> <code>`POST /waste`</code>	Site Officer, HSE Lead	Raw operational data entry for diesel, electricity, water withdrawal, hazardous waste, and safety.
factors.py <code>`/api/v1/factors`</code>	<code>`GET /emission-factors`</code> <code>`POST /emission-factors`</code> <code>`GET /unit-conversions`</code>	All (Read), Admin (Write)	Stores official emission coefficients (CEA Grid, IPCC, DEFRA) and unit ratios.
evidence.py <code>`/api/v1/evidence`</code>	<code>`POST /upload`</code> <code>`GET /{id}`</code> <code>`GET /{id}/download`</code> <code>`POST /link`</code>	Site Officer, Auditor	Cryptographic upload of utility bills, weighbridge slips, and calibration certificates with SHA-256.
consolidation.py <code>`/api/v1/consolidation`</code>	<code>`GET /group/{group_id}`</code> <code>`GET /subsidiary/{id}`</code> <code>`GET /bu/{id}`</code>	Management, Auditor	Dynamic multi-entity rollups, Scope 1/2/3 aggregation, and intensity metrics.
brsr.py <code>`/api/v1/brsr`</code>	<code>`GET /readiness`</code> <code>`GET /framework/{code}`</code> <code>`POST /answers`</code> <code>`GET /core-indicators`</code>	ESG Head, Auditor	SEBI BRSR indicator questions, readiness score calculation, and answer verification.
hr.py <code>`/api/v1/hr`</code>	<code>`GET /workforce`</code> <code>`POST /workforce`</code> <code>`GET /training`</code> <code>`POST /posh-grievance`</code>	HR Officer, HR Head	Permanent vs contract employees, gender diversity, training man-hours, and POSH cases.
hse.py <code>`/api/v1/hse`</code>	<code>`GET /incidents`</code> <code>`POST /incidents`</code> <code>`GET /inspections`</code> <code>`POST /corrective-action`</code>	HSE Lead, Site Officer	Lost Time Injury Frequency Rate (LTIFR), near-miss logs, EHS site inspections.

procurement.py `/api/v1/procurement`	`GET /suppliers` `POST /suppliers` `GET /local-sourcing-pct`	Procurement Lead	Tier-1 supply chain screening, MSME procurement share, local sourcing metrics.
governance.py `/api/v1/governance`	`GET /policies` `POST /policies` `GET /ethics-grievances`	Company Secretary, Legal	Board oversight, anti-corruption policies, whistleblower grievance tracking.
csr.py `/api/v1/csr`	`GET /projects` `POST /projects` `GET /spend-summary`	CSR Head	Schedule VII community investments, local development spend, beneficiary count.
audit.py `/api/v1/audit`	`GET /logs` `GET /logs/export` `GET /entity/{id}`	Statutory Auditor, Admin	Query immutable transaction history, tamper checks, auditor export.
reports.py `/api/v1/reports`	`POST /generate-pdf` `POST /generate-xlsx` `GET /issued`	ESG Head, Auditor	Generates statutory SEBI BRSR PDF and XLSX documents stamped with SHA-256 seal.

6. Relational Database Schema & Data Models

The persistence layer is structured across **38 SQLAlchemy ORM models** divided into 7 distinct functional domains. All tables include primary keys (UUID / deterministic string slugs), indexed foreign keys, timestamps with timezone, and audit metadata.

Domain & Tables	Key Columns & Data Types	Foreign Key Constraints & Relationships
Organizational Domain <code>`groups`</code> <code>`subsidiaries`</code> <code>`business_units`</code> <code>`projects`</code>	<code>`id`</code> (PK, String) <code>`name`</code> (String, Index) <code>`cin`</code> (String, Group only) <code>`sector`</code> / <code>`region`</code> (String) <code>`latitude`</code> , <code>`longitude`</code> (Float)	Hierarchical recursive relationships: <code>`projects.business_unit_id`</code> → <code>`business_units.id`</code> <code>`business_units.subsidiary_id`</code> → <code>`subsidiaries.id`</code> <code>`subsidiaries.group_id`</code> → <code>`groups.id`</code>
Identity & RBAC Domain <code>`users`</code> <code>`roles`</code> <code>`user_scopes`</code> <code>`permissions`</code> <code>`role_permissions`</code>	<code>`id`</code> (PK, UUID) <code>`email`</code> (Unique, Index) <code>`hashed_password`</code> (String) <code>`role_id`</code> (FK → <code>`roles.id`</code>) <code>`scope_level`</code> (GROUP / SUB / BU / PROJECT) <code>`scope_target_id`</code> (String)	Users possess exactly 1 primary role. Scopes strictly limit data access to designated organizational nodes. Permissions are mapped many-to-many via <code>`role_permissions`</code> .
Reporting & Governance <code>`reporting_periods`</code> <code>`submissions`</code> <code>`submission_versions`</code> <code>`approval_actions`</code>	<code>`id`</code> (PK) <code>`financial_year`</code> (String, e.g. '2024-25') <code>`status`</code> (DRAFT, SUBMITTED, LOCKED) <code>`submitted_by`</code> , <code>`approved_by`</code> (FK → <code>`users.id`</code>) <code>`version_number`</code> (Integer)	Submissions bind project records to an active <code>`reporting_period_id`</code> . Approval actions create an immutable chain of custody record.
Operational ESG Records <code>`fuel_records`</code> <code>`energy_records`</code> <code>`water_records`</code> <code>`waste_records`</code> <code>`safety_records`</code>	<code>`id`</code> (PK, UUID) <code>`quantity`</code> (Decimal/Float) <code>`unit`</code> (Litres, kWh, KL, MT) <code>`source_type`</code> (Grid, DG, Solar, Borewell) <code>`scope1_co2e`</code> , <code>`scope2_co2e`</code> (Float) <code>`evidence_doc_id`</code> (FK)	Every record references <code>`project_id`</code> and <code>`reporting_period_id`</code> . If <code>`evidence_doc_id`</code> is present, it links to <code>`evidence_documents.id`</code> .
Evidence Vault Domain <code>`evidence_documents`</code> <code>`evidence_links`</code> <code>`evidence_history`</code>	<code>`id`</code> (PK, UUID) <code>`file_name`</code> , <code>`storage_path`</code> (String) <code>`file_hash_sha256`</code> (64-char Hex) <code>`file_size_bytes`</code> (BigInt) <code>`mime_type`</code> (PDF, PNG, XLSX) <code>`uploaded_by`</code> (FK → <code>`users.id`</code>)	<code>`file_hash_sha256`</code> is computed upon stream arrival. <code>`evidence_links`</code> connects evidence items to specific BRSR answers or ESG records.
BRSR Compliance Framework <code>`brsr_frameworks`</code> <code>`brsr_sections`</code> <code>`brsr_indicators`</code> <code>`brsr_answers`</code> <code>`brsr_answer_sources`</code>	<code>`indicator_code`</code> (Unique, e.g. 'P6-ESS-1') <code>`principle_number`</code> (1-9) <code>`category`</code> (ESSENTIAL / LEADERSHIP) <code>`is_brsr_core`</code> (Boolean) <code>`answer_value`</code> (JSON / Text) <code>`assurance_status`</code> (Enum)	Maps SEBI question codes to answers. <code>`brsr_answer_sources`</code> records whether the value was manual input or auto-calculated from ESG records.

Audit & Assurance Domain • <code>`audit_logs`</code> • <code>`issued_reports`</code>	<code>`id`</code> (PK, BigInt) <code>`timestamp`</code> (DateTime UTC, Index) <code>`user_id`</code> , <code>`user_role`</code> (String) <code>`action`</code> (CREATE, UPDATE, APPROVE, LOCK) <code>`old_values`</code> , <code>`new_values`</code> (JSON) <code>`report_hash_sha256`</code> (IssuedReport)	Audit logs are write-only (append-only). Issued reports permanently snapshot generated PDF / Excel deliverables with a SHA-256 seal.
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7. Role-Based Access Control (RBAC) & Security Architecture

To comply with SEBI Governance Guidelines and eliminate cross-department data tampering, the portal enforces a **Dual-Vector Security Model**: Role Permissions × Organizational Scopes.

Enterprise Role	Authorized Domain	Data Entry Scope	Approval / Lock Rights
Site Officer / Field Engineer	Project Site Operations	Diesel fuel, electricity, water withdrawal, hazardous waste, concrete batching.	Submit only. Cannot approve own work.
HSE Safety Lead	Health, Safety & Environment	Lost Time Injuries, incident investigations, EHS audits, safety man-hours.	Reviews & endorses HSE records for BU.
HR / POSH Officer	Human Capital & Diversity	Workforce gender ratios, training hours, wages, POSH complaints.	Approves HR data at Subsidiary level.
BU Coordinator / Lead	Business Unit Operations	Multi-project operational reconciliation.	First-level approval of Project packages.
Group ESG Manager	Corporate Sustainability HQ	Consolidation adjustments, BRSR Section B policies, CSR spending.	Final approval before locking reporting period.
Statutory ESG Auditor	Third-Party Assurance (ICAI)	Read-Only across all 250+ sites. Sample evidence downloads.	Issues Assurance Certificates; cannot mutate raw data.
Super Administrator	IT Governance & Security	User provisioning, role assignment, emission factor updates.	Emergency lock override with mandatory audit reason.

7.1 Cryptographic Token Security & Revocation

- **JWT Algorithm:** Uses HS256/RS256 with 256-bit entropy keys stored in environment variables.
- **Token Expiration:** Access tokens expire after 480 minutes (8 working hours).
- **Instant Revocation:** An in-memory/Redis token blocklist immediately invalidates tokens upon logout, eliminating replay attacks.
- **Password Storage:** Hashed using `passlib[bcrypt]` with dynamic 12-round salt derivation.

8. Evidence Vault, SHA-256 Immutability & ICAI Audit Trail

Under ICAI Revised Assurance Standard (2024), verbal or unverified digital claims do not qualify for Reasonable Assurance. The MEIL Evidence Vault establishes an unbroken, cryptographically provable chain of custody:

- **Zero-Tamper Upload Pipeline:** When a site officer uploads a utility bill or calibration certificate, the backend computes its SHA-256 hash in real-time as bytes stream into disk storage.
- **Direct Indicator Linking:** Evidence items are linked via foreign keys to the exact BRSR indicator or energy record they substantiate, providing instant 1-click auditor verification.
- **File Integrity Verification:** Upon download or auditor inspection, the stored file's hash is recomputed and compared against `file\_hash\_sha256` in the database. Any discrepancy triggers a severe tamper alert.

- **Audit Certificate Export:** Automated generation of cryptographically signed Verification Certificates containing report hash, issued timestamp, and certifying officer credentials.

9. Deployment Architecture & Operational Runbook

The MEIL ESG platform is designed for zero-downtime containerized or on-premise execution within MEIL's private corporate infrastructure or secure sovereign cloud environments.

Step / Procedure	Command / Configuration	Verification Check
1. Environment Configuration	<code>`copy backend/.env.example backend/.env`</code> Configure database URL, JWT secret key, and CORS origins.	Ensure <code>`SECRET_KEY`</code> has at least 32 random characters.
2. Database Migration	<code>`alembic upgrade head`</code> Executes all 38 model schema definitions.	Table count matches 38; check <code>`alembic_version`</code> table.
3. Reference Data Seeding	<code>`python scripts/seed_factors_units_validation.py`</code> <code>`python scripts/seed_brsr_framework.py`</code> <code>`python scripts/seed_rbac_and_users.py`</code>	9 Principles, 140+ indicators, CEA grid factor (0.716), and corporate users seeded.
4. Backend Server Launch	<code>`uvicorn app.main:app --host 0.0.0.0 --port 8000 --workers 4`</code>	<code>`curl http://127.0.0.1:8000/health`</code> returns <code>`{"status":"healthy","database":"CONNECTED"}`</code> .
5. Frontend Build & Serve	<code>`npm run build`</code> <code>`npm run preview -- --port 5173`</code> (or Nginx reverse proxy)	HTTP 200 on <code>`http://127.0.0.1:5173/`</code> with sub-second bundle load.
6. Automated Test Suite	<code>`pytest tests/ -v`</code>	100% passing tests across emissions, consolidation, and RBAC.

ENTERPRISE TECHNICAL SPECIFICATION ENDORSEMENT

Platform Name: MEIL ESG & SEBI BRSR Enterprise Reporting Portal
Entity: Megha Engineering & Infrastructures Limited (MEIL Group HQ, Hyderabad)
System Architect: Lead Enterprise ESG Systems Architect
Compliance Reviewer: Statutory Compliance & Internal Audit Lead
Statutory Standard: SEBI BRSR Core Circulars (2021/2023/2025) & ICAI Reasonable Assurance 2024
Document Hash: SHA-256 Stamped & Cryptographically Archived in Enterprise Vault
Status: APPROVED FOR STATUTORY REPORTING & PRODUCTION OPERATIONS